

Development of Sustainable Paper, Drinking Straw and Plate Using Zea mays Husk

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Development of Sustainable Paper, Drinking Straw and Plate Using Zea mays Husk

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Abstract

Maize is a crop cultivated worldwide due to its high demand. It has been used to generate natural cellulosic fibers for the textile industry; its starch is used in preparing soups, sauces, gravies, and custards. The corn husks are often discarded or utilized to make crafts and cigarette packs. The present study focuses on extracting cellulose from the Zea mays (corn) husk by exploring different chemical methods to prepare papers and starch was also extracted from *Manihot esculenta*. The cellulose and starch obtained were analyzed for their Purity using Fourier Transform Infrared Spectroscopy (FTIR) analysis. Cellulose extracted from different methods was used to prepare paper using the screen mesh and to make a straw. Binding agents such as Starch, Agar, gelatin, Polyethylene glycol (PEG), and Sorbitol are added to the cellulose in the synthesis of bioplastics. The bioplastics were analysed for their water absorption capacity, Tensile strength and biodegradation using microorganisms. The purpose of this study is to ascertain the impact of various materials on the hardness, expansion rate of paper, straw and to improve paper strength and durability. The results reveal that agar can be used in preparing the bioplastic along with cellulose. Corn husk, most promising agricultural wastes to make sustainable and eco-friendly products like Paper and straws and plates, which reduce environmental pollution, are biodegradable and compostable.

Key words: Zea mays, Manihot esculenta, Tensile strength, Biodegradable, Sustainability.

Introduction:

Plastics, non-biodegradable compounds that are more comfortable, economical products used on a large scale in the world. It takes up hundreds of years to degrade; ingestion of marine animals leads to blockages, nutrient deficiencies, and death. They cause flooding and waterlogging in the Landscape and are toxic to mankind. Production of plastic bags leads to greenhouse gas emissions [1]. Next to plastic bags are plastic straws found in ocean clean-ups, harming marine life through ingestion and entanglement. In response, paper straws have emerged as an alternative. However, paper straws have their own set of challenges, primarily in sourcing wood pulp, which can contribute to deforestation. Paper can be produced from any natural plant, but it varies in its quality [2, 3]. In India, corn is cultivated as a food crop for both humans and animals. Corn stalks are used to make paper and wallboard; husks as filling material, woven amulets, dolls and Tamales; cobs to produce fuel, charcoal and solvents [4]. Cellulose fiber has high strength, durability, hygroscopicity, and flexibility due to its properties; they are used for making paper. It has wide application in documentation, storing information, wrapping, packaging, toweling, insulation, and photography. Cellulose, a natural polymer, forms the circular economy. It shows 3Rs such as reduce, reuse, and recycle as cellulose contributes to environmental sustainability [5]. Corn husk is an agricultural waste that poses a problem with its disposal [6]. Burning of corn husk leads to health and environmental hazards; therefore, the development of a suitable scientific and economic method of recycling of corn husk is very important [7]. The current study focuses on preparing papers using *Zea mays* husk treated with various chemicals, making drinking straws and plates. It also aims to develop a workable manufacturing procedure for producing cellulose-based drinking straws and their byproducts that are environmentally friendly.

Materials and Methods

Chemicals Required: Sodium carbonate, Hydrochloric acid, Sodium hydroxide, Potassium permanganate, Oxalic acid, and iodine, all of which were of analytical grade.

Plant sample: Corn and cassava were collected from the local market, Perambur, Tamil Nadu, India. The samples were authenticated as *Zea mays* (Husk) and *Manihot esculenta* (pulp) by a Dr.R.Muralidharan, Associate Professor & Head, Department of Botany, DDGDVC.

Extraction of Cellulose:

Corn husk is chopped into small pieces and the methods used to process are

Method I: Boiling with Sodium carbonate for 5-6 hours,

Method II: Boiling with Sodium bicarbonate for 5 hours,

Method III: Boiling with 0.1N Hydrochloric acid for 3 hours, followed by 0.1N Sodium hydroxide treatment for 3 hours.

After following each method pulp obtained was thoroughly washed with tap water, drained, and macerated using a blender. To get a pure white paper, the pulp obtained was further treated with potassium permanganate and oxalic acid, each for 3 to 4 hours. After treatment, pulp was thoroughly washed in tap water until it reached a pH of 7.

Preparation of Paper:

The cellulose pulp was made into a slurry using distilled water and transferred into a tub. The mould and deckle are dipped, transferred, and water is pressed using a sponge gently and then squeezed out the water using a blotter sheet. The resulting paper sheet can be removed from the mesh screen mould while it is still wet, or it may be removed after it is completely dry. Sandwich the paper between an absorbent material and dry firmly with a flat, heavy object.

Extraction of Starch:

2 Kg *Manihot esculenta* (Cassava) Collected from the Local market. The skin was peeled off, pulp was cut into pieces, and soaked in tap water for 2-3 hours. Samples were macerated, filtered with the help of a muslin cloth. Filtrate was left undisturbed for 5-6 hours. Starch gets sedimented at the bottom of the vessel. Filtrate was transferred onto the plate and kept under the sun for 2 days for drying. Dried starch was made into a fine powder using a blender & stored at room temperature for future use.

Qualitative Analysis:

1. **Chlor-Zinc-Iodine Test:** Mix zinc chloride, potassium iodide, and iodine in water. Place a small amount of the cellulose-containing material in a test tube or on a slide. Add a few drops of chlor-zinc-iodine solution to the sample. The violet or blue-violet colour formation indicates the presence of cellulose.

2. **Iodine Test:** The starch obtained from the *Manihot esculenta* was dissolved in distilled water, added a few drops of Iodine solution were added and boiled for 10 minutes.

FTIR Analysis:

The extracted cellulose and starch samples were analyzed using FTIR spectroscopy to determine their purity, to identify chemical constituents and elucidate compound structures. The powder samples were analyzed using a Shimadzu Fourier transform infrared spectrophotometer (FTIR 8400S Shimadzu, Japan). FTIR spectra were recorded for all experimental runs.

Synthesis of Bioplastic:

The extracted cellulose was treated with various binding agents, including Starch, Agar, Gelatin, Polyethylene glycol, and Sorbitol at different concentrations to synthesize bioplastics. The extracted cellulose pulp (20 g) was weighed and mixed with 2% starch, 2% Agar, 2% sorbitol, 2% polyethylene glycol and 2% Sorbitol separately and then boiled for 15 min at 100 °C to prepare a bioplastic polymer. The bioplastics were analyzed for tensile strength and water absorption capacity and biodegradation using natural soil.

Characterization of Tensile Strength:

After the production of cellulose paper and bioplastic, the tensile test was conducted to determine the mechanical properties. All the samples were cut with dimensions of 100 mm x 30 mm to assess the ability of the film to resist applied pressure. Finally, all data were calculated. The graph was plotted to compare the results of the paper and different types of bioplastics.

Water Absorption Capacity:

This test is designed to determine the water absorption behavior of cellulose paper and bioplastics. To test, all the samples were cut into 50 mm x 50 mm dimensions and pre-weighed, then placed in a petri dish, and 25 ml of distilled water was added. The weight of the film is noted periodically at the 1st hour, 2nd hour, 3rd hour and 24th hour. Water absorption capacity of the film was determined in triplicate and calculated using the following formula:

$$\% \text{ Water absorption capacity} = [(W_t - W_o) / W_o] \times 100$$

Whereas,

Wt (g) is the weight of the bioplastic at the time being

Wo (g) is the initial weight of bioplastic before being immersed in distilled water.

Biodegradation Analysis:

Cellulose paper and Bioplastics prepared using Starch, Agar, gelatin, PEG, and Sorbitol, of 5 g each, were weighed and used for the analysis. All the samples were buried for 15 days in natural soil, and the weight of the Cellulose paper and bioplastic was measured after 5-day intervals.

Results and Discussion:

Extraction of Cellulose and Starch:

The treatment with 0.1 N Hydrochloric acid and 0.1 N Sodium hydroxide resulted in complete digestion of the corn husk, whereas treatment with sodium carbonate and sodium bicarbonate was less effective. The cellulose pulp was further used for making paper sheets (**Fig.1,2**). Starch extracted from 1 kg of *Manihot esculenta* was found to be 800 gm was stored at room temperature for further use (**Fig.3**). Starch, a storage polysaccharide found in grains, tubers, roots, and unripe fruit that is widely used in food and industry due to its easily available, biodegradable, renewable and safe [8]. Cellulose extracted from the Corn husks of 75 % and snake plant fiber of 25 % produces a quality paper with significant texture, colour, and thickness [9].

Qualitative analysis:

The Chlor-Zinc-Iodine Test shows that the extracted pulp from *Zea mays* husk contains cellulose and the Iodine Test indicates the presence of starch obtained from the *Manihot esculenta* (**Fig.4**). The presence of cellulose and starch in *Zea mays* husk and *Manihot esculenta*, respectively, has significant implications for their potential use as alternative sources of these biomolecules. Cellulose can be used in paper production, textiles, and bioplastics [10], while starch can be used in food, pharmaceuticals, and biodegradable plastics [11].

Fig 1. Extraction of Cellulose from Zea mays



Collection of Corn Husk



Corn Husk were Chopped



Corn Husk treated with Sodium carbonate



Corn Husk treated with Sodium Bicarbonate



Corn Husk treated with 1N Hydrochloric acid

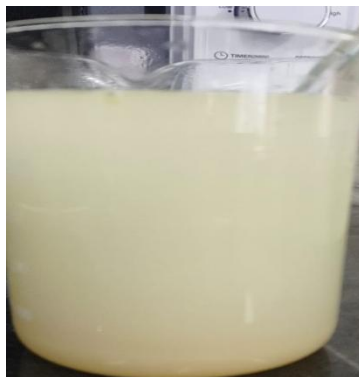


Corn Husk treated with 1N Sodium Hydroxide

Fig 2. Decolourization of Extracted Cellulose from Zea mays Husk



Cellulose treated with
Potassium permanganate



Cellulose treated with Decolourizing Agent

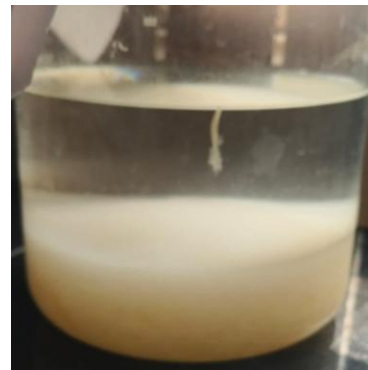


Fig 3. Extraction of Starch from *Manihot esculenta*



Manihot esculenta
sample



Peel removed from
Manihot esculenta



Manihot esculenta Chopped into
pieces and soaked in water



Macerated *Manihot esculenta* was
kept for 5 to 6 hours



Powdered Cassava starch

Fig 4. Iodine test for Starch

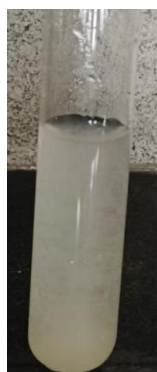


Control - Starch



Starch treated with Iodine

Fig 5. Chlor-Zinc-Iodine Test for Cellulose



Control - Cellulose



Hydrolyzed Cellulose was treated with Chlor-Zinc-Iodine

Fig 6. Measurement of pH for the extracted Cellulose



FTIR Analysis:

The extracted cellulose and starch were analyzed for their purity by FTIR analysis. Peak obtained from cellulose extracted from corn husk shows the presence of Hydroxyl groups (-OH) [12], Alkyl groups (-CH₂- or -CH₃) [13], Unsaturated compounds or aromatic rings, Ether or ester groups, Amines or Amides and Aromatic compounds [14, 15], which confirms the sample is cellulose and hemicellulose (**Fig.7,8; Table 1,2**). This analysis provides valuable information on the molecular structure and composition of the cellulose and starch samples.

Fig 7. FTIR Analysis of Cellulose

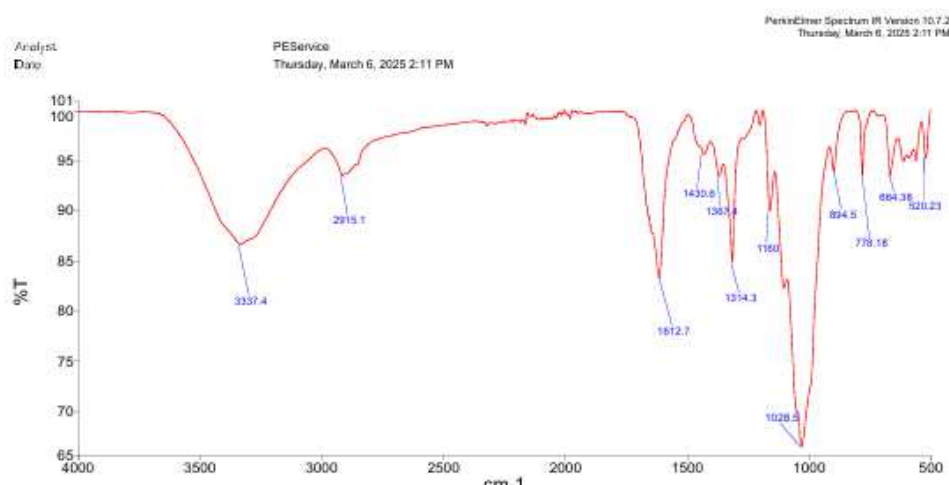


Fig 8. FTIR Analysis of Starch

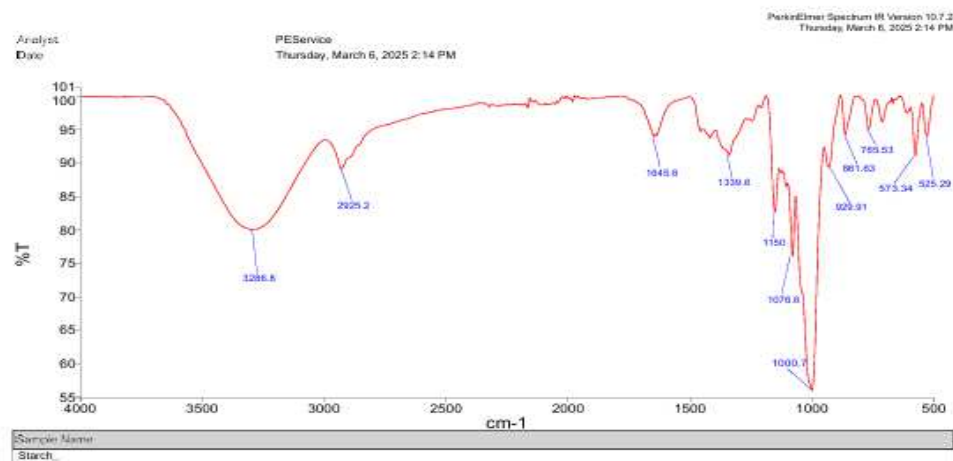


Table 1. FTIR Analysis of Cellulose extracted from *Zea mays* Husk

BOND	TYPE	FREQUENCY RANGE cm ⁻¹	NATURE
C=C	Alkenes	1610-1680	Variable
C-H	Alkanes	1340-1470	Strong
C-H	Alkenes	675-995	Strong
C-H	Alkanes	2850-3000	Stretching
O-H	Alcohol	3584-3700	Stretching
C-H or C-N	Amines or Amides	1314.3	Bending or stretching
C-O	Hydroxyl groups	1028.5	Stretching
C-C or C-O	Alkyl or ether	520.23	Stretching or bending

Table 2. FTIR Analysis of Starch Extracted from *Manihot esculenta*

BOND	TYPE	FREQUENCY RANGE cm ⁻¹	NATURE
O-H	Hydrogen bonding Alcohol's, Phenols	3200-3600	Variable, sometimes broad
C-H	Alkanes	1340-1470	Strong
C-O	Alcohols, ethers, carboxylic acids, esters	1050-1300	Strong
C-H	Alkenes	675-995	Strong
C-H	Alkanes	2850-3000	Stretching
C-H	Alkenes	675-995	Strong
C-H	Alkenes	675-995	Strong
C-C or C-O	Alkyl groups or Ether	500- 573.34	Bending or Vibration

Product Formulation:

The cellulose pulp and starch were used to prepare papers, plates and cups (**Fig.9.a, b**). The prepared paper was used to make straws. The undigested residues from the sodium carbonate and sodium bicarbonate treatment were further used to make mats for kids and dishwash scrubbers (**Fig.10. a, b**). The tensile strength of paper used for making straw and paper plates was also analysed (**Fig. 11, Table 3,4**). The straws were analysed for biodegradation using natural soil over 15 days to ensure environmental sustainability of the product. After 5 days, the weight of the sample was reduced to 3.5 g, on the 10th day to 2.5 g and on the 15th day it was completely precipitated. The flavonoid and carotenoid components in potato and cassava peels protect cells from oxidative damage and enhance immunity. The nutritionally significant components in the potato peel can be used in the food and pharmaceutical industries [16, 17]. Paper straws have a higher water absorption rate than the polylactic acid straws but it has less biodegradation in natural soil of < 0.3% [18]. The cellulose was found to be higher in cassava peels than that of cocoa pod husks and lower than *Pinus pinaster* and *Eucalyptus globulus* (wood). Lignin content is also low in cocoa pod husks and wood plants, which may be an alternative source that can be applied in papermaking industries [19].

Fig 9a. Products Prepared using Synthesized Cellulose



Paper Plate



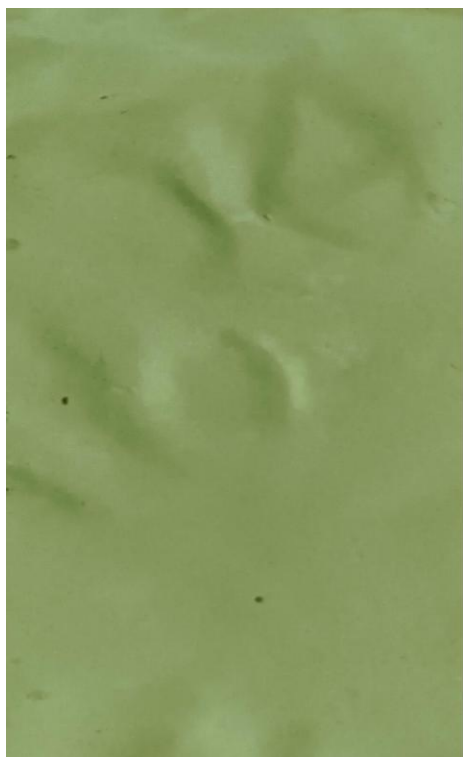
Circular Paper



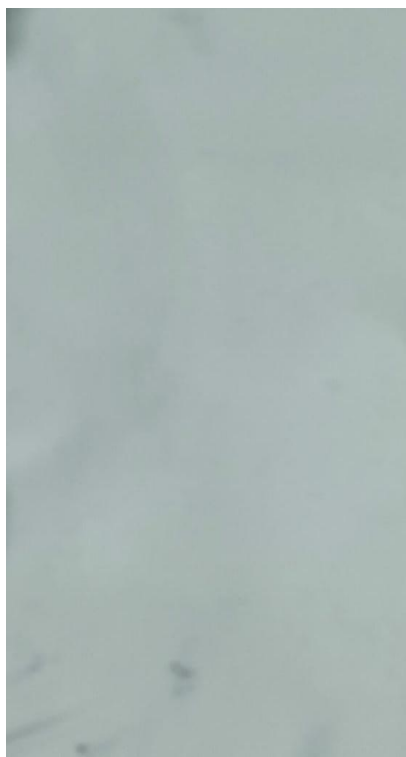
Paper Cup



Fig 9b. Products Prepared using Synthesized Cellulose



Paper Prepared using Cellulose



Paper Prepared using Decolourized Cellulose



Straw Prepared using Cellulose



Straw Prepared using Bleached Cellulose

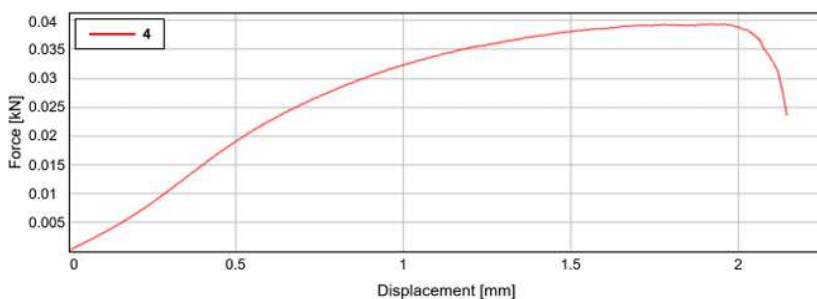
Fig 10a. Mat Prepared using Undigested Cellulose and Starch



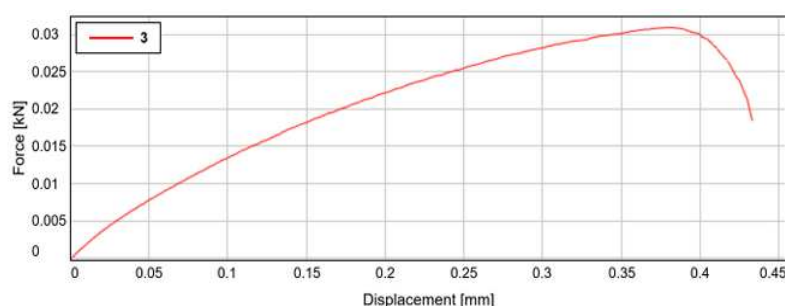
Fig 10b. Dish wash Scrubber Prepared using Undigested Cellulose and Starch



Fig 11. Tensile strength of Cellulose



Cellulose Unbleached used to make paper cup



Bleached Cellulose paper used to make straw

Preparation of Bioplastic and its characterization:

The extracted cellulose was treated with binding agents, such as 2% agar, 2% sorbitol, 2% Polyethylene glycol, and starch, at different concentrations to synthesize the bioplastic (Fig. 12). The synthesized bioplastic was analyzed for its tensile strength and water absorption capacity. Cassava peel can be used to make so many different products and since it is so rich in starch, researchers are becoming increasingly keen on it [20]. The ratio of amylopectin to amylase in the extracted starch was 5:1, leading to gelatinization and enhancing solubility [21]. The cellulose extracted from corncob using hot sodium hydroxide and hydrogen peroxide bleaching stabilizes the mechanical properties of poly lactic acid [22].

The Tensile strength of biofilm prepared using agar exhibits the highest tensile strength among all the tested specimens, indicating its potential for applications requiring high strength and durability (**Fig. 13, Table 3,4**). The water absorption capacity of the bioplastic shows that agar can be used as a binding agent since its water absorption rate is found to be low among the other

binding agents, such as Polyethylene glycol, starch and sorbitol exhibit poor performance. The biodegradability of the prepared bioplastics was analyzed for 15 days and the samples were completely digested using natural soil (**Fig.13**). Thus, bioplastics ensure the environmental sustainability. Bioplastic synthesized using rice straw shows mechanical resistance, tensile strength with break at 45 MPa and 10 Mpa indicates high mechanical performance both in dry and wet status that shows both the Industrial and environmental impact in carbon reduction and sustainable bioplastic [23]. Glycerol combines with starch act as a good bioplastic material. Fillers such as chitosan, clay, and ZnO. and glycerol, plasticizers exhibit different characters and gives opposite contribution to the quality of tensile strength and Elongation [24]. Bio films synthesized from potato starch by blending with glycerol and citric acid showed better efficiency and durability. Increased glycerol reduces film strength but enhances flexibility and degradation rate. Depending on the concentration of starch exhibits different pore size that influence the water absorption capacity of the film properties. Films degrade faster with more glycerol within 14 days [25]. Non-wood biomass is a promising alternative fiber resources for pulp and paper production. An alternative plastic carry bags can be synthesized from polysaccharides by blending with plasticizers to ensure sustainable environment.

Table 3. Tensile strength of Biofilm prepared using Cellulose Extracted from *Zea mays* Husk.

S.No.	Specimen	Maximum Force [N]	Tensile stress at Tensile strength [MPa]	Tensile strain (Displacement) at Break (Standard) [%]	Tensile stress at Break (Standard)[MPa]
1.	Cellulose unbleached paper cup	39.44	10.43	8.57	6.42
2.	Cellulose Paper used to make straw	30.94	2.95	1.44	1.83
3.	Cellulose blended with Agar	57.98	15.34	5.73	9.60
4.	Cellulose blended with PEG	1.90	-0.38	6.53	-0.48
5.	Cellulose blended with Sorbitol	2.17	0.31	11.26	0.02
6.	Cellulose blended with starch, Agar, gelatin	47.94	10.65	1.41	10.30
7.	Cellulose blended with Starch	30.64	4.54	1.18	2.79

Table 4. Water Absorption Capacity of Biofilm Prepared Using Cellulose Extracted from *Zea mays* Husk

S.No.	Specimen	% Of Water Absorption for 24 hours
1.	Cellulose blended with Agar	14
2.	Cellulose blended with PEG	19.7
3.	Cellulose blended with Sorbitol	33.8
4.	Cellulose paper cup	45

Fig 12. Bioplastic formed using Binding agents



Cellulose blended with PEG



Cellulose blended with Sorbitol



Cellulose blended with Agar



Cellulose blended with Starch, Gelatin & Agar

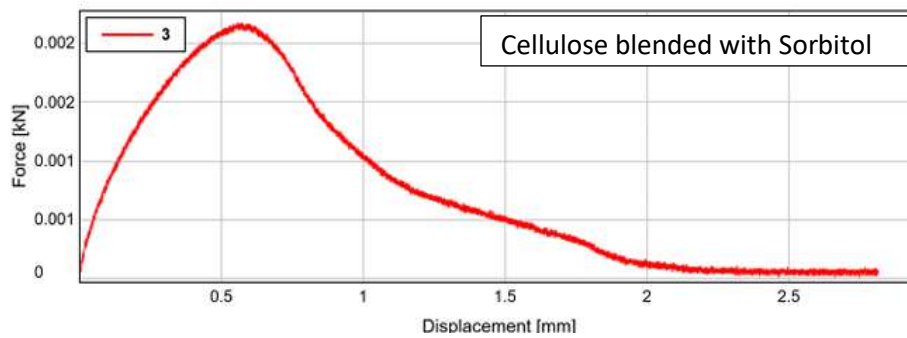
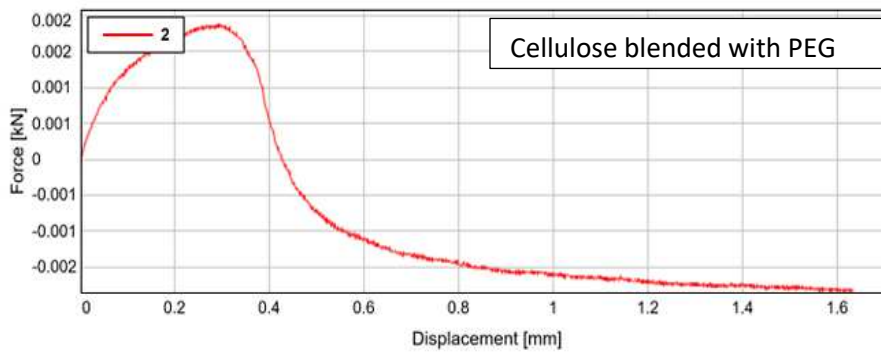
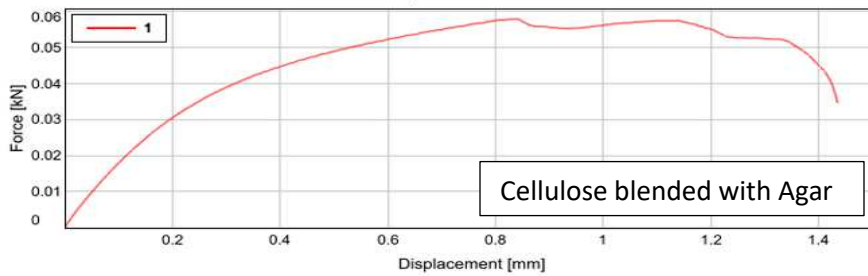


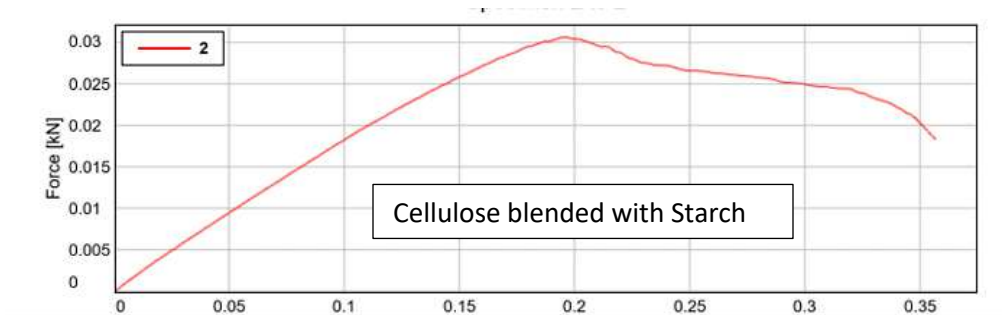
Cellulose blended with Starch



Cellulose blended with Gelatin

Fig 13. Tensile strength of Bioplastic formed using Cellulose





Conclusion:

Paper is a multipurpose product all over the world that leads to rapid deforestation. To resolve by using non-timber resources for making paper. The study reveals that corn husk is a renewable resource and abundant agricultural waste, reducing reliance on wood pulp. Production of cellulose straws from corn husk is a low-cost raw material, reducing production expenses and benefiting the local farming communities. They serve as a renewable source, eco-friendly and in the formulation of biodegradable products and reduce the usage of plastic.

Abbreviations

FTIR - Fourier Transform Infrared Spectroscopy

PEG - Polyethylene glycol

mm- Millimeter

%- Percentage

g- Gram

°C- Degree Celsius

FTIR - Fourier Transform Infrared Spectrophotometer

Kg - Kilogram

< - Less than

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Declarations:

Ethics, Consent to Participate, and Consent to Publish declarations: not applicable.

Clinical trial number: not applicable.'

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Data Availability:

The datasets generated during study on Development of Sustainable Paper, Drinking Straw and Plate Using *Zea mays* Husk analysed during the current study are available from the corresponding author on reasonable request.

Ethics and consent to participate:

The plant material used in this study consisted of edible corn and cassava purchased from a local market, Perambur, Tamil Nadu, India. No samples were collected from the wild, and therefore no specific permissions or licences were required. All procedures complied with local and national guidelines for research involving commercially available plant materials. The plant identity as *Zea mays* (husk) and *Manihot esculenta* (pulp) was confirmed by a Dr.R.Muralidharan, Associate Professor & Head, Department of Botany, DDGDVC.